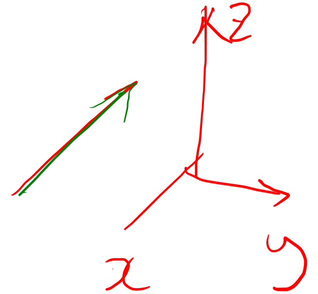
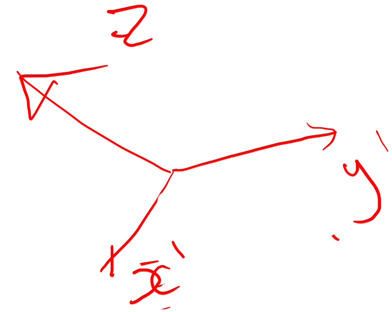
$\Phi \qquad\qquad \Phi$

$u_i \longrightarrow \underline{u}$ vector

$A_{ij} \qquad \underline{\underline{A}}$

$B_{ijk} \qquad \underline{\underline{\underline{B}}}$





$$\underline{\underline{A}} \cdot \underline{u} = \underline{b}$$

$$A_{ij} u_j = b_i \qquad i = 1 \ldots 3 \quad j = 1 \ldots 3$$

3D

Tensor de orden en un problema de dimensión 2

$i = 1 \ldots 2$

$$\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \qquad \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$$

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \qquad \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}$$

$v = \begin{bmatrix} u_1^2 \\ u_2^2 \\ u_3^2 \end{bmatrix}$

$u_i^2 = b$

ind. libre

$$\begin{cases} u_1^2 = b \\ u_2^2 = b \\ u_3^2 = b \end{cases}$$

$i = 1, 2, 3$

$$\boxed{u_j^2 = b}$$

escalar

$u_i \cdot u_i = b \Rightarrow u_1 \cdot u_1 + u_2 \cdot u_2 + u_3 \cdot u_3 = b$

$u_1^2 + u_2^2 + u_3^2 = b$

$\|\underline{u}\|^2$

ind dummy

$u_j \cdot u_j$

orden 1 → 1 indice libre

$\to b_j A_{jk} \equiv v_k \equiv b_l A_{lk} = v_k$

Problema de 3 dimensiones: $i, j = 1, 2, 3$

$$\delta_{ii} = 3$$

$$\delta_{ij} \Rightarrow \underline{\underline{I}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\delta_{11} = 1$

$\delta_{12} = 0$

$\vdots$

A_{ii}

$A_{ii}^{2} = A_{11} + A_{22} + A_{33}$

$$A_{ii} = \sum_{i=1}^{3} A_{(i)(i)} =$$

$A_{ijk} \rightarrow$ rango 3

$A_{ijkl} \rightarrow$ rango 4

$A_{ij} K_j$

$A_{ij} = \delta_{ij}$

$$A_{ii} = \delta_{ii} = \overset{1}{\delta_{11}} + \overset{1}{\delta_{22}} + \overset{1}{\delta_{33}}$$

$= 3$

$i = 1,2,3$

$$\delta_{ij}\delta_{ij} = 3$$

$$\delta_{ij}\delta_{ij} = \delta_{ii} = 3$$

$$\underbrace{\delta_{ij} \cdot A_{ik}}_{\text{dummy}} = A_{jk}$$

libre

libre

$$\boxed{\delta_{ij} u_j = u_i}$$

$$\begin{cases} \delta_{1j} u_j = \delta_{11} \cdot u_1 + \cancel{\delta_{12}}^{0} \cdot u_2 + \cancel{\delta_{13}}^{0} u_3 = \delta_{11} \cdot u_1 = u_1 \\ \delta_{2j} u_j = \cancel{\delta_{21}}^{0} u_1 + \delta_{22} \cdot u_2 + \cancel{\delta_{23}}^{0} \cdot u_3 = \delta_{22} \cdot u_2 = u_2 \\ \delta_{3j} u_j = \cancel{\delta_{31}}^{0} \cdot u_1 + \cancel{\delta_{32}}^{0} \cdot u_2 + \delta_{33} u_3 = \delta_{33} \cdot u_3 = u_3 \end{cases}$$

$$\text{equivalentes}\begin{cases} u_{ij} b_j = c_i & \longrightarrow \quad 3 \text{ ecuaciones} \\ u_{ij} b_j - c_i = 0 & \longrightarrow \quad 3 \text{ ecuaciones} \\ u_{ik} b_k - c_i = 0 & \longrightarrow \quad 3 \text{ ecuaciones} \end{cases}$$

$$e_{ijk}e_{jki} = 6$$

$A_{jkpq} =$ dummy

$= e_{ijk}\, e_{ipq} = \delta_{jp}\delta_{kq} - \delta_{jq}\delta_{kp}$

libres

$\delta_{ij} = \delta_{ji}$

$e_{ijk} = e_{kij} = -e_{ikj}$

321

312

e_{kji}

e_{ijk}

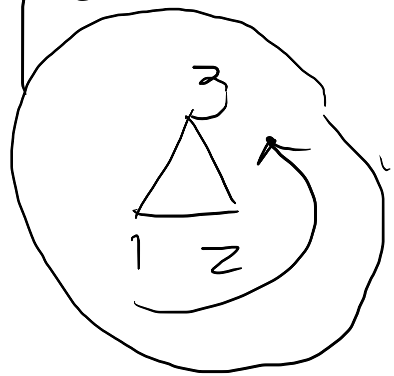
The permutation symbol, sometimes called the **Levi-Civita symbol**,

112 … 122 … $\Rightarrow 0$

123 312 231 $\Rightarrow 1$

213 132 321 $\Rightarrow -1$

Tensor de 3er orden



σχολιο

$$\bigcirc = e_{ijk}e_{jki} = 6$$

3D

$$e_{ijk}\cdot e_{ijk} = \delta_{jj}\cdot \underbrace{\delta_{kk}}_{\delta_{pp}} - \overbrace{\delta_{kj}\delta_{jk}}^{\delta_{kk}}$$

$\cancel{\delta_{jj}}$

$$= 3\cdot 3 - 3 = 6$$

$$e_{ijk}A_jA_k = 0$$

$$\ldots + A_2 A_3 e_{123} + \ldots$$

$$\ldots + e_{123} \cdot A_2 \cdot A_3 + \ldots$$

$$\ldots + e_{123} \cdot A_3 \cdot A_2 + \ldots$$

$$2 e_{ijk} A_j A_k$$

$$e_{ijk} A_j A_k = \frac{1}{2}\left(e_{ijk} A_j A_k + e_{ijk} A_j A_k\right)$$

cambio de posición

intercambio los indices jk de e_{ijk}

$$= \frac{1}{2}\left(e_{ijk} A_j A_k - e_{ikj} A_j A_k\right)$$

conmutación de un producto

$$= \frac{1}{2}\left(e_{ijk} A_j A_k - e_{ikj} A_k A_j\right)$$

renombre índices dummy

$$= \frac{1}{2}\left(e_{ijk} A_j A_k - e_{ijk} A_j A_k\right) = 0$$

$$e_{ipq} A_p A_q$$

$$e_{ijk} A_j A_k$$

$$e_{ikj} A_k A_j$$

$$e_{ijk} A_j A_k = -e_{ikj} A_j A_k = -e_{ikj} A_k A_j$$

renombro k con j
j con k

$$\underbrace{e_{ijk} A_j A_k}_{a_i} = -\underbrace{e_{ijk} A_j A_k}_{a_i}$$

$$\varnothing = -\varnothing = 0$$

$$a_i = -a_i$$

$$a_1 = -a_1 \Rightarrow a_1 = 0$$

$$a_2 = -a_2 \Rightarrow a_2 = 0$$

$$a_3 = -a_3 \Rightarrow a_3 = 0$$

$$b = -b$$

$$2b = 0$$

$$b = 0/2 = 0$$

$$a_i = 0$$

$$e_{ijk} A_j A_k = 0$$

$$\delta_{ij}\delta_{jk} = \delta_{ik}$$

libre

$\Rightarrow \delta_{ij}\delta_{jk} = \delta_{ik}$

$j = i$

$$\delta_{ij}e_{ijk} = 0$$

$e_{iik} = 0$